

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Scenario-Based Research Assessment

COURSE NAME: ARTIFICIAL INTELLIGENCE AND EXPERT SYSTEMS COURSE CODE: MLA01

COURSE OUTCOME COVERED

CO1: Apply search algorithms and heuristic techniques to solve state-space search and optimization problems. (BTL 3)

Assessment Tool Weightage: 45 %

Total Marks: 50

Time: 60 Mins

No. of Questions: 5

Annexure A

Scenario-Based Research Assessment

Code of Conduct:

I Sk. Fawaz Ahamad / 192425390 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criterion	Weightage	Marks Obtained
Identification of Latest AI Application	10%	
Problem Analysis and Domain Understanding	15%	
AI Technologies and Working Mechanism	20%	
Research Quality and Literature Review	10%	
Real-World Implementation and Case Analysis	10%	
Impact, Challenges, and Future Scope	15%	
Originality and Critical Thinking	10%	
Organization, Presentation, and Documentation	10%	
Total	100%	

Signature of the Faculty

Total Marks Awarded

ANSWER ANY ONE

No.	AI Domain	Scenario-Based Research Question
1	Healthcare	Healthcare organizations are increasingly adopting Artificial Intelligence to improve diagnosis, treatment, and patient monitoring. Identify and research a latest AI-based application in healthcare (such as AI-assisted medical diagnosis, AI radiology systems, robotic surgery, AI drug discovery, or virtual healthcare assistants). Analyze its working principles, AI techniques used, real-world implementation, benefits, and limitations.
2	Finance	Financial institutions are leveraging Artificial Intelligence to enhance security, decision-making, and customer services. Identify and research a latest AI application in the finance domain (such as AI fraud detection systems, AI-powered financial advisors, algorithmic trading platforms, or intelligent banking assistants). Discuss its functionality, technologies involved, advantages, and challenges.
3	Retail	Retail companies are adopting AI-driven solutions to provide personalized and intelligent shopping experiences. Identify and research a latest AI application in retail (such as Generative AI shopping assistants, AI recommendation engines, virtual try-on systems, or AI-based demand forecasting). Explain its application, AI methods used, impact on customers and businesses, and future scope.
4	Autonomous Vehicles	The transportation sector is rapidly evolving with AI-based autonomous systems. Identify and research a latest AI application in autonomous vehicles (such as self-driving technology, AI-based perception systems, autonomous delivery vehicles, or intelligent traffic management). Analyze the AI technologies involved, real-world deployment, benefits, and challenges.
5	Robotics	Intelligent robots are being developed for applications in industries, healthcare, education, and daily life. Identify and research a latest AI application in robotics (such as humanoid robots, collaborative robots, service robots, or AI-powered warehouse robots). Explain their capabilities, AI techniques, real-world usage, and future developments.
6	Manufacturing	Manufacturing industries are integrating AI to achieve smart and automated production environments. Identify and research a latest AI application in smart manufacturing (such as AI-based predictive maintenance, computer vision quality inspection, digital twins, or autonomous manufacturing systems). Discuss its working, industrial applications, advantages, and limitations.

Structure of the Report:

S.No	Report Component	Student Deliverable / Guiding Question
1	Title of AI Application	Provide the name of the latest AI application selected and the corresponding AI domain.
2	Domain Selection	Identify the application domain (Healthcare / Finance / Retail / Autonomous Vehicles / Robotics / Manufacturing).
3	Introduction and Background	Describe the AI application and provide an overview of its purpose and importance.
4	Problem Identification	Explain the real-world problem or challenge addressed by this AI application.
5	Need for AI Solution	Analyze why AI is required and how it improves traditional approaches.
6	AI Technologies Used	Identify the AI techniques involved (Machine Learning, Deep Learning, NLP, Computer Vision, Generative AI, Robotics, etc.).
7	Working Mechanism	Explain how the AI system works, including input data, processing methods, decision-making, and output generation.
8	Data Sources and Processing	Describe the type of data used and how the data is processed for intelligent decision-making.
9	Real-World Implementation	Identify organizations, industries, or real-world scenarios where this AI application is currently implemented.
10	Impact and Benefits	Evaluate the improvements achieved through the AI application in terms of efficiency, accuracy, cost, user experience, or automation.
11	Challenges and Limitations	Discuss technical, ethical, privacy, security, and deployment challenges associated with the application.
12	Future Scope and Emerging Trends	Explain possible enhancements, future developments, and research opportunities in this AI application.
13	Conclusion	Summarize the significance of the AI application and its contribution to solving real-world problems.
14	References	Provide research papers, technical articles, industry reports, and reliable sources referred for the study.

Rubrics for Evaluation

S.No	Criteria	Weightage (%)	Excellent (100–90%)	Good (89–75%)	Satisfactory (74–60%)	Improvement (59–50%)
1.	Identification of Latest AI Application	10%	Selects a highly relevant, emerging AI application with strong justification.	Selects a recent and relevant AI application.	Selects a suitable application with limited justification.	Application is outdated or irrelevant.
2.	Problem Analysis and Domain Understanding	15%	Clearly explains the real-world problem, domain context, and need for AI with critical analysis.	Good explanation with adequate analysis.	Basic explanation with limited analysis.	Problem statement is unclear or incomplete.
3.	AI Technologies and Working Mechanism	20%	Accurately explains AI techniques, algorithms, architecture, and workflow with technical depth.	Good explanation with minor omissions.	Covers only basic AI concepts.	AI technologies are poorly explained or incorrect.
4.	Research Quality and Literature Review	10%	Uses recent, credible sources with proper citations and excellent synthesis.	Good quality references with appropriate citations.	Limited references with basic discussion.	Few or unreliable references; poor citation.
5.	Real-World Implementation and Case Analysis	10%	Thoroughly analyzes industrial implementation with appropriate examples.	Good implementation analysis.	Basic implementation discussion.	Minimal or no real-world analysis.
6.	Impact, Challenges, and Future Scope	15%	Critically evaluates benefits, limitations, ethical issues, and future trends.	Good evaluation with minor gaps.	Basic discussion of impact and future scope.	Limited evaluation or missing sections.
7.	Originality and Critical Thinking	10%	Demonstrates independent research, innovation, and critical evaluation.	Good analytical thinking.	Some originality with limited analysis.	Mostly descriptive or copied content.
8.	Organization, Presentation, and Documentation	10%	Exceptionally organized, professional presentation, error-free, and properly referenced.	Well organized with minor issues.	Adequately organized with some errors.	Poor organization and documentation.

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES (SIMATS)

School of Engineering

Department of Artificial Intelligence & Machine Learning

AI Applications in Finance: Credit Card Fraud Detection Using Artificial Intelligence

Submitted in partial fulfillment of the requirements for

MLA0103 – Artificial Intelligence and Expert Systems

Name	Sk. Fawaz Ahamad
Register Number	192425390
Department	B.Tech Artificial Intelligence & Machine Learning
Year	III Year
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Submission Date	14/07/2026

Chennai, Tamil Nadu

Academic Year: 2024–2028

1. Selected AI Application

AI-Based Credit Card Fraud Detection System

The AI application selected for this assignment is the **AI-Based Credit Card Fraud Detection System** used in the **Finance** domain.

Credit card fraud detection is one of the most successful and widely adopted Artificial Intelligence applications in modern banking. Every day, banks process millions of financial transactions. Detecting fraudulent transactions manually is almost impossible because of the huge volume of data. Artificial Intelligence solves this problem by continuously monitoring customer transactions, identifying unusual spending patterns, and predicting fraudulent activities in real time.

Unlike traditional rule-based systems, AI continuously learns from historical transaction data and adapts to new fraud patterns. This enables financial institutions to protect customer accounts, minimize financial losses, and improve the overall security of digital payment systems.

1.1 Why This Application Was Chosen

This application was selected because it is one of the most practical and widely implemented AI solutions in the finance industry. The reasons for selecting this application are listed below.

- **Real-World Application**

Almost every bank, digital payment platform, and financial institution uses AI-based fraud detection systems to protect customer accounts and online transactions.

- **High Business Impact**

Financial fraud causes billions of dollars in losses every year. AI significantly reduces these losses by detecting suspicious transactions before they are completed.

- **Easy to Understand**

The working process can be explained using everyday examples such as online shopping, ATM withdrawals, and credit card payments.

- **Covers Multiple AI Concepts**

The application demonstrates important Artificial Intelligence concepts including:

- Machine Learning
- Deep Learning
- Big Data Analytics
- Pattern Recognition
- Predictive Analytics
- Data Mining

2. Any One Scenario (Real-World Use Case)

To understand the AI application clearly, let us consider a real-world scenario in the Finance domain. The selected AI application is AI-Based Credit Card Fraud Detection System.

Scenario

A customer named Rahul regularly uses his credit card in Chennai for grocery shopping, online purchases, restaurant payments, and fuel expenses. His normal transaction amount ranges between ₹500 and ₹5,000.

One day, while Rahul is in Chennai, a transaction of ₹1,25,000 is attempted from New York, USA using his credit card. Since this transaction is completely different from Rahul's normal spending behaviour, the AI-based fraud detection system immediately identifies it as suspicious.

The AI system automatically analyzes:

- Previous transaction history
- Transaction amount
- Transaction location
- Transaction time
- Device and IP address
- Merchant details

Within a few seconds, the machine learning model predicts that the transaction is fraudulent. The bank instantly blocks the transaction and sends an OTP verification and security alert to Rahul's registered mobile number. Rahul confirms that he did not perform the transaction, preventing financial loss.

2.1 Example

Parameter	Normal Activity	Suspicious Transaction
Location	Chennai, India	New York, USA
Amount	₹500 – ₹5,000	₹1,25,000
Transaction Time	Morning / Evening	Midnight
Device	Rahul's Mobile	Unknown Device
Risk Level	Low	Very High

3. Why This AI Application?

Artificial Intelligence has become an essential technology in the finance industry because millions of financial transactions are performed every day through credit cards, debit cards, mobile banking, and online payment systems. Detecting fraudulent activities manually is difficult, time-consuming, and prone to human error. The **AI-Based Credit Card Fraud Detection System** solves this problem by automatically analyzing customer transactions in real time and identifying suspicious activities before financial losses occur.

Unlike traditional fraud detection systems that rely on fixed rules, AI continuously learns from historical

transaction data and adapts to new fraud patterns. Machine Learning algorithms analyze customer spending behaviour, transaction amount, location, merchant details, transaction time, device information, and previous purchase history to determine whether a transaction is genuine or fraudulent. This enables banks to make faster and more accurate decisions while improving customer security.

3.1 What Problems Does AI Solve?

- Detects fraudulent transactions in real time.
- Prevents financial losses for customers and banks.
- Reduces manual verification and operational costs.
- Identifies unusual customer spending behaviour.
- Improves the accuracy of fraud detection.
- Provides 24×7 continuous transaction monitoring.

3.2 Why Is AI Better Than Traditional Methods?

Traditional Method	AI-Based Method
Manual verification	Fully automated fraud detection
Fixed security rules	Learns from transaction data
Slow decision making	Real-time prediction
Lower accuracy	High prediction accuracy
Difficult to detect new fraud patterns	Continuously adapts to new fraud techniques

3.3 Why Is AI Needed in Finance?

- Enhances digital payment security.
- Protects customer financial information.

4. Scenario Explanation (How the System Works)

This section explains how the **AI-Based Credit Card Fraud Detection System** works whenever a customer performs a financial transaction. The complete process consists of seven simple stages, from transaction initiation to fraud detection and customer notification.

Step 1 – Customer Initiates a Transaction

The customer performs an online or offline payment using a credit card through a shopping website, ATM, Point of Sale (POS) machine, or mobile banking application.

Step 2 – Transaction Data Collection

The banking system collects essential transaction details such as transaction amount, customer location, merchant information, device ID, IP address, transaction time, and previous transaction history.

Step 3 – Data Processing

The collected transaction data is stored securely and preprocessed. Missing values are removed, useful features are extracted, and the data is prepared for machine learning analysis.

Step 4 – AI-Based Analysis

The Machine Learning model compares the current transaction with the customer's historical spending behaviour and identifies unusual patterns that may indicate fraudulent activity.

Step 5 – Fraud Prediction

The AI model classifies the transaction as either **Genuine** or **Fraudulent** based on its prediction score and risk analysis.

Step 6 – Alert and Verification

If the transaction is identified as suspicious, the system immediately blocks or temporarily holds the transaction and sends an OTP verification or security alert to the customer's registered mobile number or banking application.

Step 7 – Final Decision

After customer verification, the transaction is either approved or permanently blocked. Every completed transaction is stored in the database and becomes new training data, enabling the AI model to continuously improve its fraud detection accuracy.

5. Architecture of the AI System

The architecture of the **AI-Based Credit Card Fraud Detection System** consists of multiple layers that work together to identify fraudulent financial transactions in real time. The system collects customer transaction data, processes it using Artificial Intelligence and Machine Learning algorithms, and provides an instant decision on whether the transaction is genuine or fraudulent.

The architecture is divided into four major stages: **Input, Processing, AI Model, and Output**. Each stage performs a specific function to ensure accurate fraud detection while maintaining customer security and transaction reliability.

5.1 Input

The input layer collects all transaction-related information required for fraud detection. The collected data includes:

- Customer transaction history
- Credit card details
- Transaction amount
- Transaction location
- Merchant information
- Device ID and IP address
- Date and time of transaction

This information is received through banking applications, ATMs, Point of Sale (POS) terminals, and online payment gateways.

5.2 Processing

The processing layer stores and prepares the collected data for analysis. During this stage, the system:

- Cleans and validates transaction data.
- Removes duplicate or missing values.
- Extracts important transaction features.
- Converts raw data into a format suitable for Machine Learning.

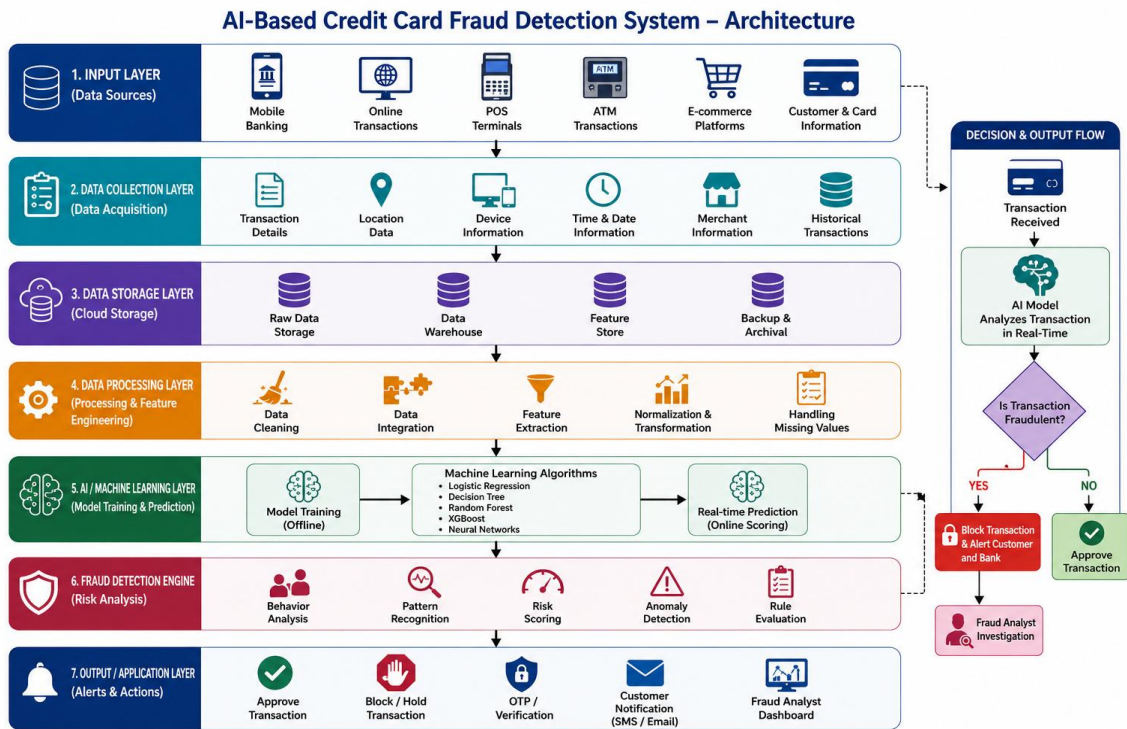
This improves the accuracy and efficiency of fraud detection.

5.3 AI Model

The AI Model is the core of the system. Machine Learning algorithms analyze transaction patterns and compare current transactions with historical customer behaviour.

Common algorithms used include:

- Decision Tree
- Random Forest
- Logistic Regression
- Neural Networks



6. Components Used and Their Purpose

The AI-Based Credit Card Fraud Detection System consists of several important components that work together to detect fraudulent transactions efficiently.

S.No	Component	Purpose
1	Customer / User	Performs financial transactions.
2	Banking Application	Receives transactions and displays alerts.
3	Data Collection	Collects transaction details.
4	Database	Stores customer and transaction data.
5	Data Processing	Cleans and prepares transaction data.
6	AI / ML Model	Predicts fraudulent transactions.
7	Fraud Detection Engine	Calculates fraud risk score.
8	Alert System	Sends OTP and fraud notifications.
9	Fraud Dashboard	Displays fraud reports for bank officials.

Summary: - These components work together to provide real-time fraud detection, improve transaction security, reduce financial losses, and protect customers from unauthorized transactions.